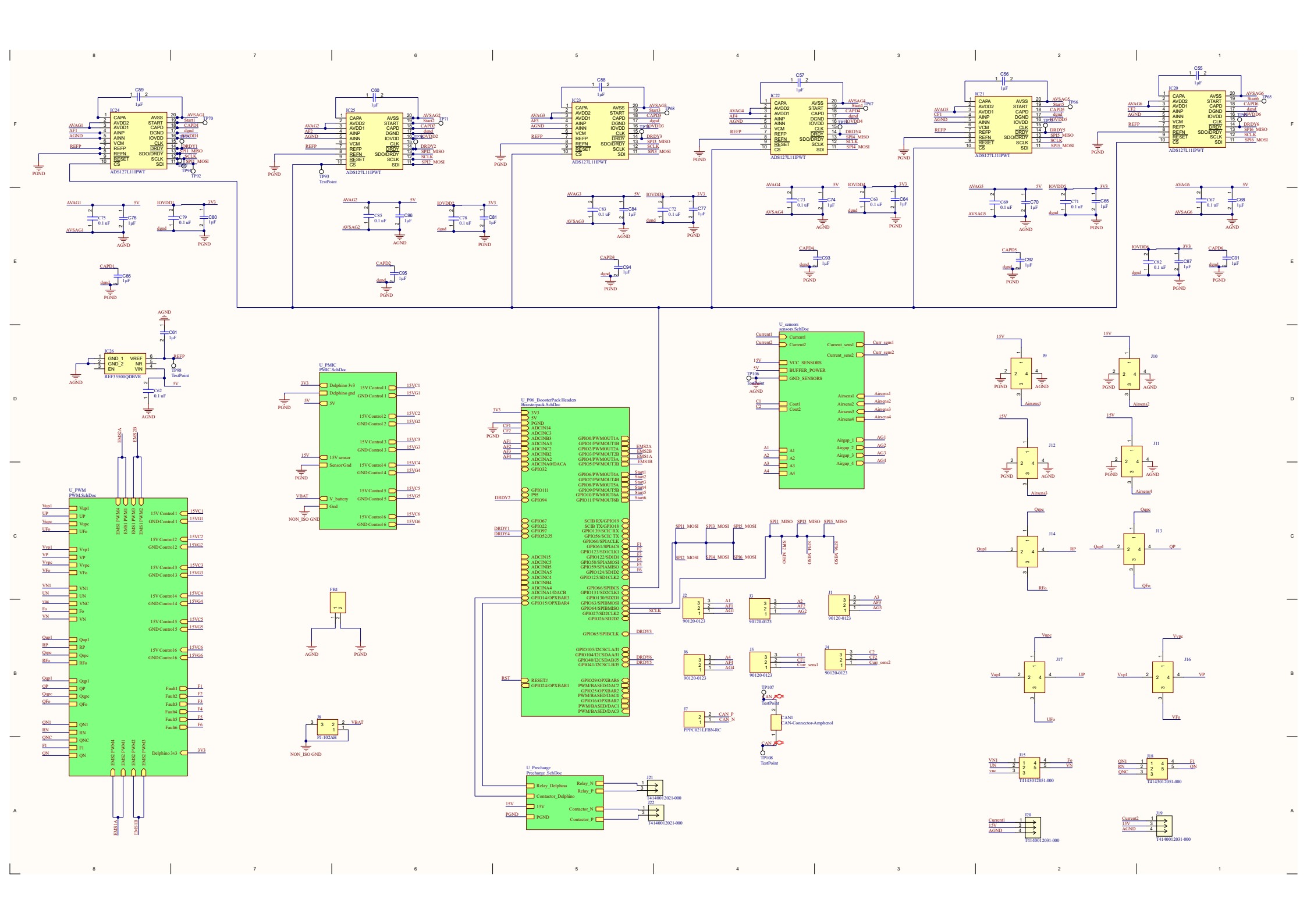
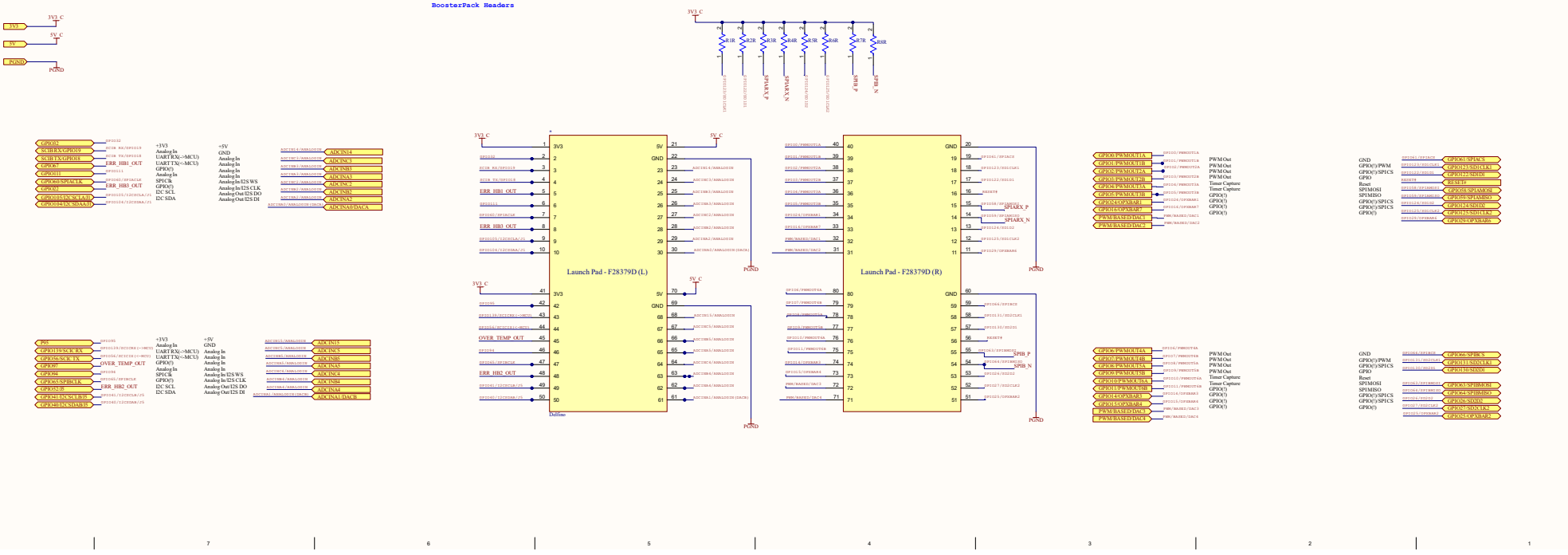
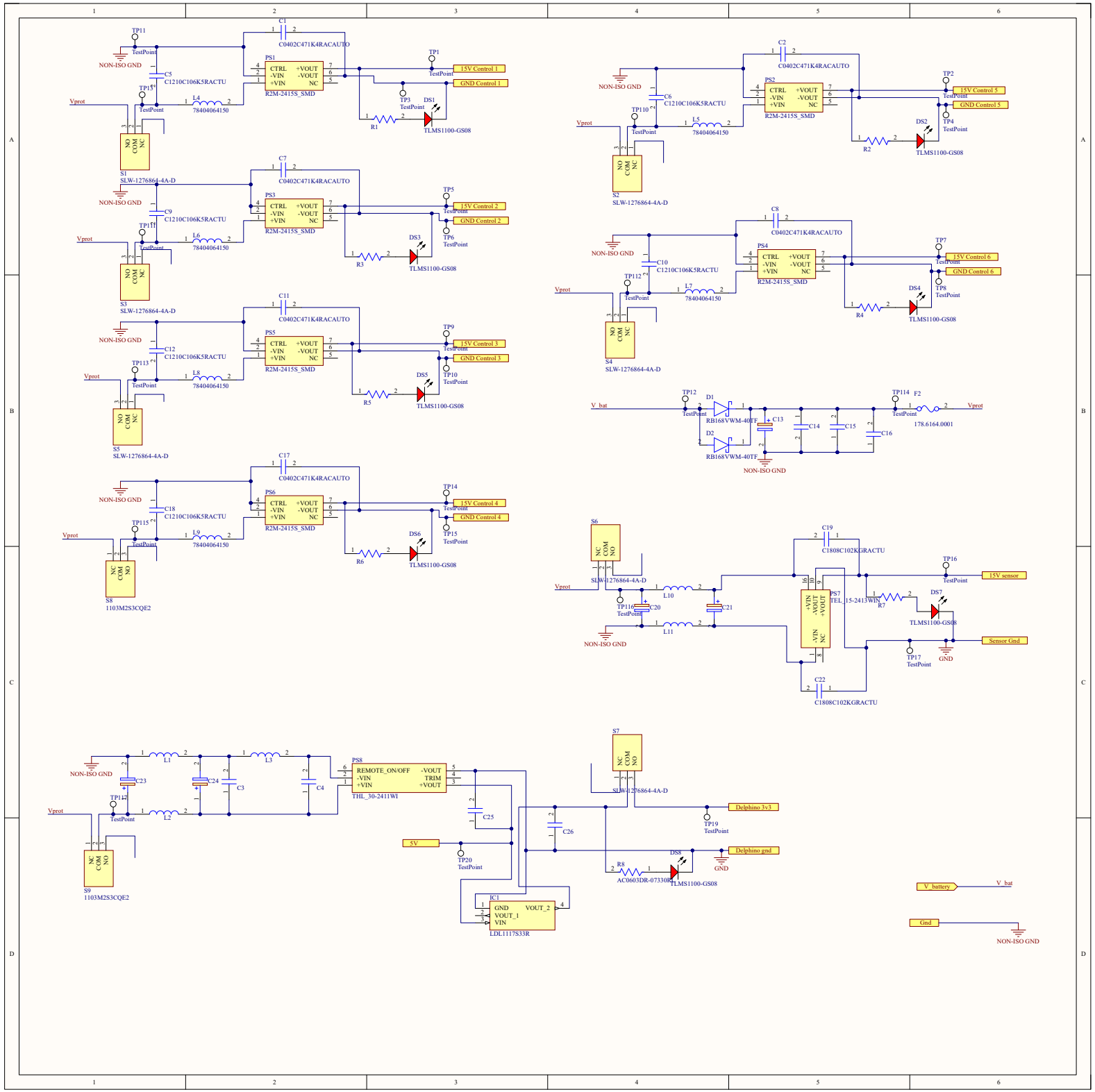
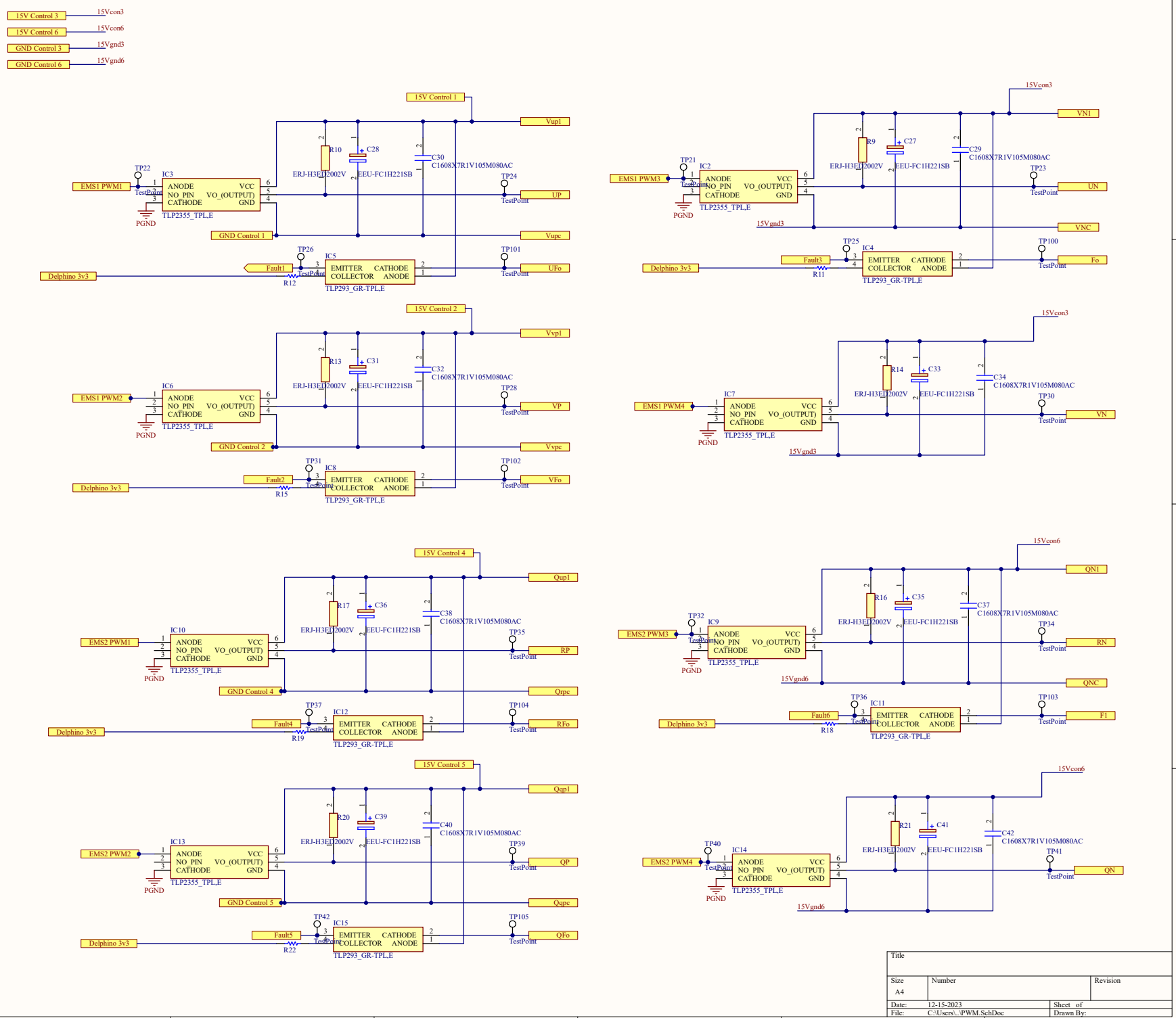
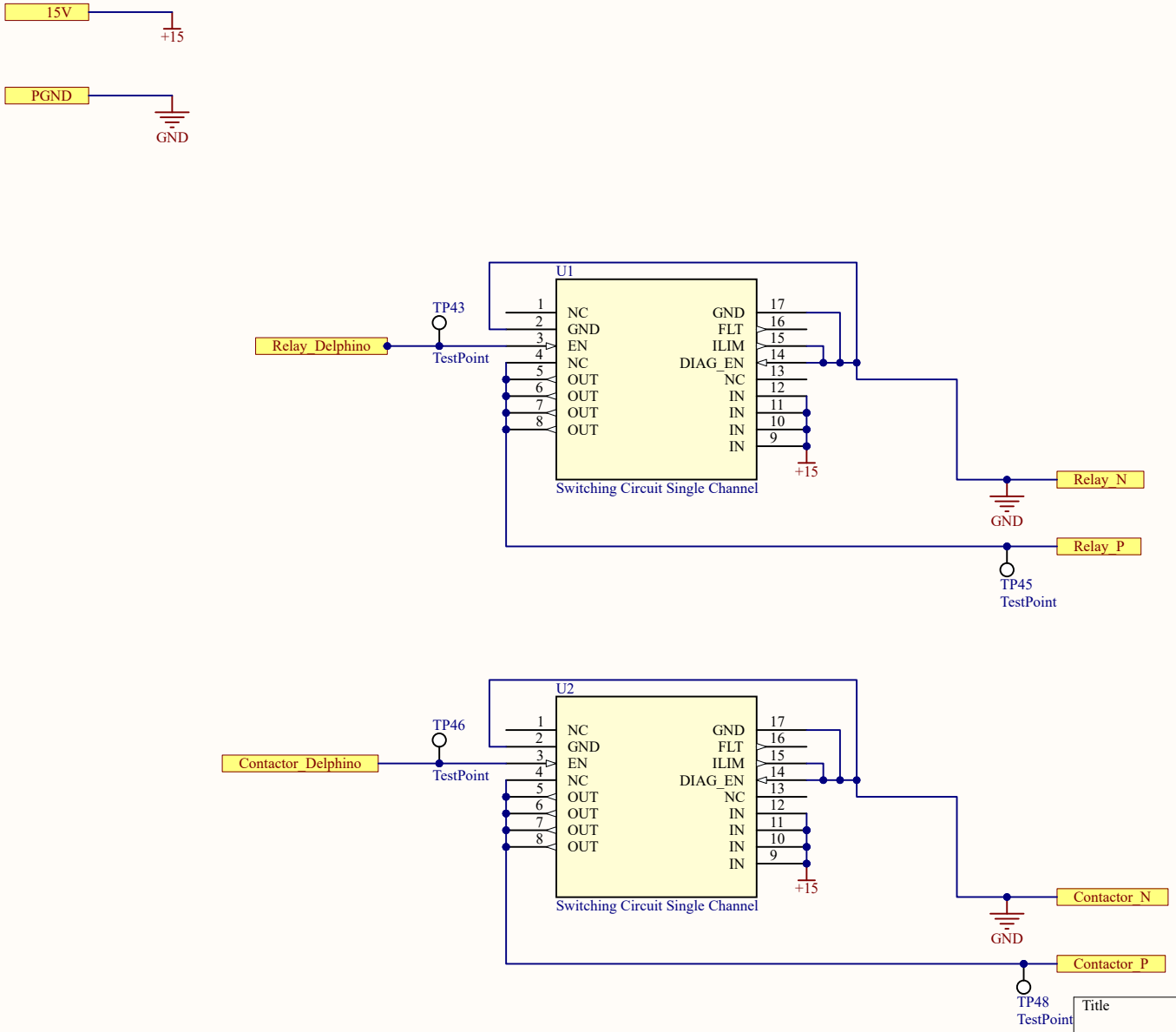
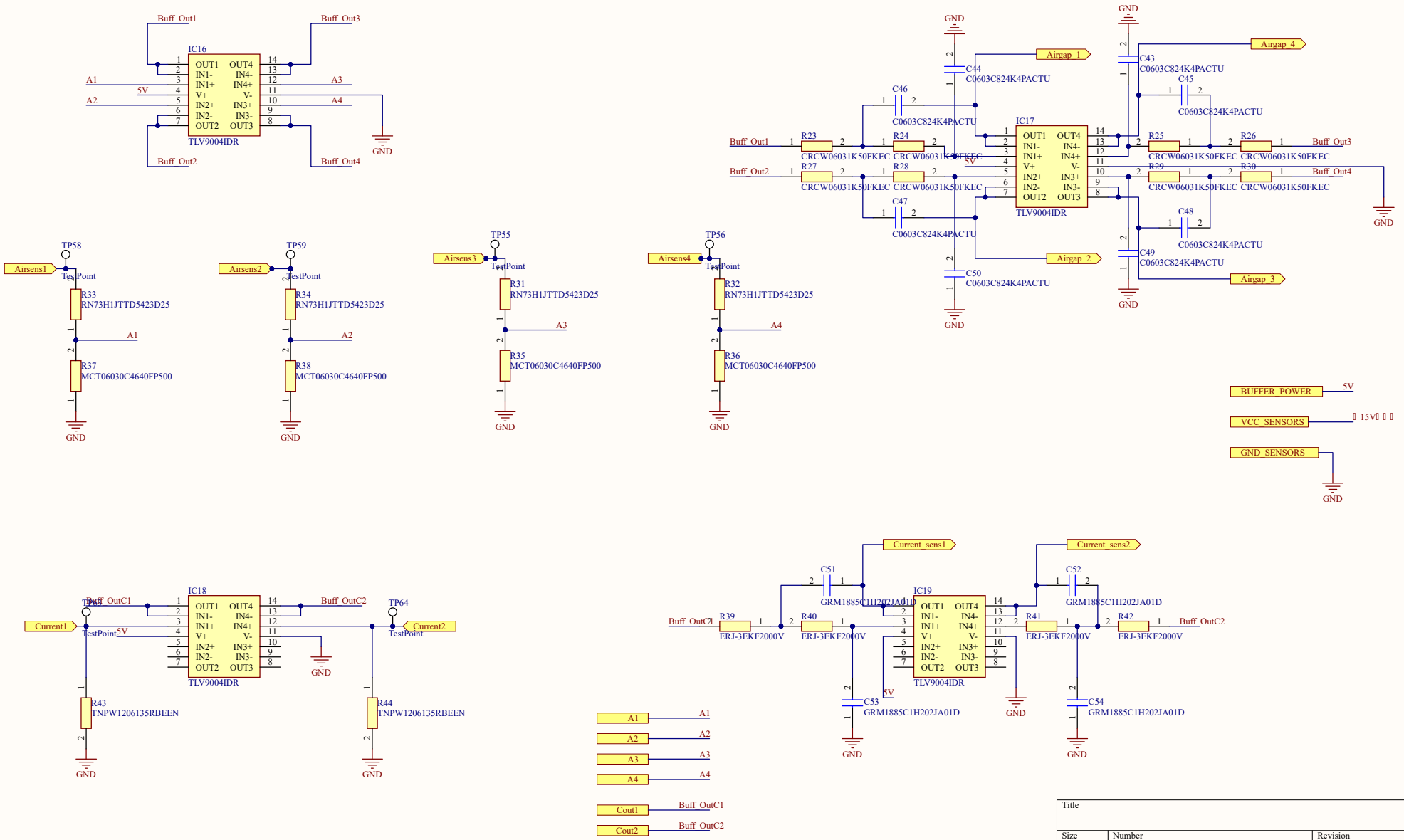

U_PWM
PWM.SchDoc
U_PMIC
PMIC.SchDoc
U_P06_BoosterPack Headers
Boosterpack.SchDoc
U_sensors
sensors.SchDoc
U_Precharge
Precharge_SchDoc
ADS127L111IPWT
REF35500QDBVR
CAN1
CAN-Connector-Amphenol
PPPC021LFBN-RC
90120-0123
T4140012021-000
T4140012031-000
T4143012051-000
PJ-102AH
NON_ISO GND


BoosterPack Headers
3V3_C
5V_C
3V3
5V
PGND
R1R
R2R
R3R
R4R
R5R
R6R
R7R
R8R
SPIA_N
SPIA_P
SPIB_P
SPIB_N
Launch Pad - F28379D (L)
Launch Pad - F28379D (R)
Define
SPIARX_P
SPIARX_N
SPIB_P
SPIB_N
GPIO34
GPIO87
GPIO111
GPIO60/SPICLK
GPIO22
GPIO41/I2CSCLB
GPIO40/I2CSDAB
+3V3
Analog In
UART RX(->MCU)
UART TX(<-MCU)
GPIO(!)
Analog In
SPICS
Analog In/I2S WS
Analog In/I2S CLK
I2C SCL
I2C SDA
+5V
GND
Analog In
Analog In
Analog In
Analog In
Analog In/I2S WS
Analog In/I2S CLK
Analog Out/I2S DO
Analog Out/I2S DI
ADCINA4/DACA
ADCINA2
ADCINB2
ADCINA3
ADCINB3
ADCINA5
ADCINB4
ADCINC2
ADCINC3
ADCINC4
ADCINB5
ADCINA1
ADCINB1
ADCINC1
ADCINC5
ADCINB6
ADCINA6
ADCINB14
ADCINA14
GPIO61/SPICS
GPIO22/SDIDI
RESET
GPIO58/SPIMOSI
GPIO59/SPIMISO
GPIO24/SDIDT
GPIO23/SDICLK
GPIO29/OPXBAR6
GPIO6/PWMOUT4A
GPIO7/PWMOUT4B
GPIO8/PWMOUT5A
GPIO9/PWMOUT5B
GPIO10/PWMOUT6A
GPIO11/PWMOUT6B
GPIO14/OPXBAR3
GPIO15/OPXBAR4
PWM Out
Timer Capture
GPIO(!)
GPIO(!)
GPIO(!)
GPIO(!)
GPIO63/SPIMOSI
GPIO64/SPIMISO
GPIO26/SDIDT
GPIO27/SDICLK
GPIO25/OPXBAR5
GPIO66/SPICS
GPIO31/SDICLK
GPIO30/SDIDI


TP11
TestPoint
NON-ISO GND
C1
C0402C471K4RACAUTO
C5
C1210C106K5RACTU
PS1
CTRL +VOUT
-VIN -VOUT
+VIN NC
R2M-2415S_SMD
TP1
TestPoint
15V Control 1
GND Control 1
TP13
TestPoint
Vprot
L4
78404064150
TP3
TestPoint
DS1
R1
TLMS1100-GS08
NO COM NC
S1
SLW-1276864-4A-D
C2
C0402C471K4RACAUTO
NON-ISO GND
C6
C1210C106K5RACTU
PS2
CTRL +VOUT
-VIN -VOUT
+VIN NC
R2M-2415S_SMD
TP2
TestPoint
15V Control 5
GND Control 5
TP110
TestPoint
Vprot
L5
78404064150
TP4
TestPoint
DS2
R2
TLMS1100-GS08
NO COM NC
S2
SLW-1276864-4A-D
C7
C0402C471K4RACAUTO
NON-ISO GND
C9
C1210C106K5RACTU
PS3
CTRL +VOUT
-VIN -VOUT
+VIN NC
R2M-2415S_SMD
TP5
TestPoint
15V Control 2
GND Control 2
TP111
TestPoint
Vprot
L6
78404064150
TP6
TestPoint
DS3
R3
TLMS1100-GS08
NO COM NC
S3
SLW-1276864-4A-D
C8
C0402C471K4RACAUTO
NON-ISO GND
C10
C1210C106K5RACTU
PS4
CTRL +VOUT
-VIN -VOUT
+VIN NC
R2M-2415S_SMD
TP7
TestPoint
15V Control 6
GND Control 6
TP112
TestPoint
Vprot
L7
78404064150
TP8
TestPoint
DS4
R4
TLMS1100-GS08
NO COM NC
S4
SLW-1276864-4A-D
C11
C0402C471K4RACAUTO
NON-ISO GND
C12
C1210C106K5RACTU
PS5
CTRL +VOUT
-VIN -VOUT
+VIN NC
R2M-2415S_SMD
TP9
TestPoint
15V Control 3
GND Control 3
TP113
TestPoint
Vprot
L8
78404064150
TP10
TestPoint
DS5
R5
TLMS1100-GS08
NO COM NC
S5
SLW-1276864-4A-D
V_bat
TP12
TestPoint
D1
RB168VWM-40TF
D2
RB168VWM-40TF
C13
C14
C15
C16
NON-ISO GND
TP114
TestPoint
F2
178.6164.0001
Vprot
C17
C0402C471K4RACAUTO
NON-ISO GND
C18
C1210C106K5RACTU
PS6
CTRL +VOUT
-VIN -VOUT
+VIN NC
R2M-2415S_SMD
TP14
TestPoint
15V Control 4
GND Control 4
TP115
TestPoint
Vprot
L9
78404064150
TP15
TestPoint
DS6
R6
TLMS1100-GS08
NC COM NO
S8
1103M2S3CQE2
S6
NC COM NO
SLW-1276864-4A-D
Vprot
L10
L11
C19
C1808C102KGRACTU
TP116
TestPoint
C20
C21
NON-ISO GND
PS7
TEL_15-2413WIN
+VIN +VOUT
-VIN NC
TP16
TestPoint
15V sensor
R7
DS7
TLMS1100-GS08
Sensor Gnd
TP17
TestPoint
GND
C22
C1808C102KGRACTU
NON-ISO GND
L1
L3
C23
C24
C3
C4
PS8
REMOTE_ON/OFF -VOUT
-VIN TRIM
+VIN +VOUT
THL_30-2411WI
Vprot
TP117
TestPoint
L2
NC COM NO
S9
1103M2S3CQE2
S7
NC COM NO
SLW-1276864-4A-D
C25
C26
5V
TP20
TestPoint
Delphino 3v3
TP19
TestPoint
Delphino gnd
GND
R8
AC0603DR-07330RL
DS8
TLMS1100-GS08
IC1
GND VOUT_2
VOUT_1
VIN
LDL1117S33R
V_battery
V_bat
Gnd
NON-ISO GND


15V Control 3
15Vcon3
15V Control 6
15Vcon6
GND Control 3
15Vgnd3
GND Control 6
15Vgnd6
15V Control 1
Vup1
TP22
IC3
EMS1 PWM1
ANODE
NO_PIN
CATHODE
VCC
VO_(OUTPUT)
GND
TLP2355_TPL,E
PGND
R10
ERJ-H3ED2002V
C28
EEU-FC1H221SB
C30
C1608X7R1V105M080AC
TP24
UP
GND Control 1
Vupc
TP26
IC5
Fault1
R12
Delphino 3v3
EMITTER
COLLECTOR
CATHODE
ANODE
TLP293_GR-TPL,E
TP101
UFo
15V Control 2
Vvp1
TP28
IC6
EMS1 PWM2
R13
C31
C32
VP
GND Control 2
Vvpc
TP31
IC8
Fault2
R15
TP102
VFo
15V Control 4
Qup1
TP35
IC10
EMS2 PWM1
R17
C36
C38
RP
GND Control 4
Qupc
TP37
IC12
Fault4
R19
TP104
RFo
15V Control 5
Qqp1
TP39
IC13
EMS2 PWM2
R20
C39
C40
QP
GND Control 5
Qqpc
TP42
IC15
Fault5
R22
TP105
QFo
TP21
IC2
EMS1 PWM3
R9
C27
C29
VN1
TP23
UN
VNC
TP25
IC4
Fault3
R11
TP100
Fo
IC7
EMS1 PWM4
R14
C33
C34
TP30
VN
TP32
IC9
EMS2 PWM3
R16
C35
C37
TP34
RN
QNC
TP36
IC11
Fault6
R18
TP103
F1
IC14
EMS2 PWM4
R21
C41
C42
TP41
QN
Title
Size
A4
Number
Revision
Date:
12-15-2023
Sheet of
File:
C:\Users\..\PWM.SchDoc
Drawn By:


15V
+15
PGND
GND
U1
TP43
Relay_Delphino
TestPoint
1 NC
2 GND
3 EN
4 NC
5 OUT
6 OUT
7 OUT
8 OUT
17 GND
16 FLT
15 ILIM
14 DIAG_EN
13 NC
12 IN
11 IN
10 IN
9 IN
+15
Switching Circuit Single Channel
Relay_N
GND
Relay_P
TP45
TestPoint
U2
TP46
Contactor_Delphino
TestPoint
1 NC
2 GND
3 EN
4 NC
5 OUT
6 OUT
7 OUT
8 OUT
17 GND
16 FLT
15 ILIM
14 DIAG_EN
13 NC
12 IN
11 IN
10 IN
9 IN
+15
Switching Circuit Single Channel
Contactor_N
GND
Contactor_P
TP48
TestPoint
Title
Size
A4
Number
Revision
Date: 12-15-2023
Sheet of
File: C:\Users\..\Precharge .SchDoc
Drawn By:


IC16
TLV9004IDR
Buff_Out1
Buff_Out2
Buff_Out3
Buff_Out4
A1
A2
A3
A4
5V
GND
IC17
TLV9004IDR
Airgap_1
Airgap_2
Airgap_3
Airgap_4
C44 C0603C824K4PACTU
C46 C0603C824K4PACTU
C47 C0603C824K4PACTU
C50 C0603C824K4PACTU
C43 C0603C824K4PACTU
C45 C0603C824K4PACTU
C48 C0603C824K4PACTU
C49 C0603C824K4PACTU
R23 CRCW06031K50FKEC
R24 CRCW06031K50FKEC
R27 CRCW06031K50FKEC
R28 CRCW06031K50FKEC
R25 CRCW06031K50FKEC
R26 CRCW06031K50FKEC
R29 CRCW06031K50FKEC
R30 CRCW06031K50FKEC
TP58
TP59
TP55
TP56
Airsens1
Airsens2
Airsens3
Airsens4
TestPoint
R33 RN73H1JTTD5423D25
R34 RN73H1JTTD5423D25
R31 RN73H1JTTD5423D25
R32 RN73H1JTTD5423D25
R37 MCT06030C4640FP500
R38 MCT06030C4640FP500
R35 MCT06030C4640FP500
R36 MCT06030C4640FP500
BUFFER_POWER
VCC_SENSORS
GND_SENSORS
15V
IC18
TLV9004IDR
Buff_OutC1
Buff_OutC2
Current1
Current2
TP63
TP64
R43 TNPW1206135RBEEN
R44 TNPW1206135RBEEN
IC19
TLV9004IDR
Current_sens1
Current_sens2
C51 GRM1885C1H202JA01D
C52 GRM1885C1H202JA01D
C53 GRM1885C1H202JA01D
C54 GRM1885C1H202JA01D
R39 ERJ-3EKF2000V
R40 ERJ-3EKF2000V
R41 ERJ-3EKF2000V
R42 ERJ-3EKF2000V
Cout1
Cout2
Title
Size
A4
Number
Revision
Date: 12-15-2023
File: C:\Users\..\sensors.SchDoc
Sheet of
Drawn By: